UMKC SCHOOL OF SCIENCE & ENGINEERING · HACK-A-ROO 2026

OptiSched: Uncertainty-Aware Enrollment Forecasting with CP-SAT Course Scheduling

Calibrating Per-Course Conformal Demand Bounds from 19 Historical Registrar Terms to Reduce Week-One Seat Shortages

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OPEN-SOURCE BENCHMARK
Code & FOIA Replay
github.com/JoeDoan/CS5551_Advanced-Software-Engineering-

−61% **Unseated Student Seats**
153 → 60 seats vs. point forecast · −82% vs. prior (338 → 60)

2 / 40 **Courses Over Room Capacity**
Down from 9 courses with point forecast and 8 with prior-term copy

34 / 40 **Held-Out Courses Inside 80% Bound**
85.0% empirical coverage · 39 / 40 (97.5%) inside 95% interval

3 / 3 **Counterfactual What-If Re-Solves**
Best possible schedule found, 0 room or instructor conflicts

01 PROBLEM & HOW OPTISCHED DIFFERS — 19/40 Short

University classrooms are assigned months before registration opens. When actual enrollment exceeds room capacity, students face week-one seat shortages and departments must relocate sections. On held-out Spring 2026 (**40 STEM courses**), a point forecast under-predicted **19 of 40 courses**.

How OptiSched Differs: Existing academic schedulers size rooms from prior-term headcount or a single point forecast. OptiSched combines per-course calibrated uncertainty bounds with a constraint programming solver (**CP-SAT**), cutting unseated student seats by **61%** vs. point forecasts (153 → 60) and **82%** vs. prior-term copy (338 → 60).

VS. TYPICAL APPROACHES	PRIOR TERM	POINT	OPTISCHED
Training history	1 term	19 terms	19 terms
Demand intervals	None	None	80% & 95%
Room buffer rule	Flat +5%	Flat +5%	80% bound
Faculty preferences	Manual edits	Manual edits	3-tier (84% match)

02 END-TO-END SYSTEM ARCHITECTURE — 5-Stage Pipeline

UIUC Freedom of Information Act (FOIA) Registrar Data — STAGE 1 · CORPUS
22,991 sections → 800 course-terms · 40 STEM courses · 19 terms (Fall 2016 to Fall 2025)

↓ 19-term enrollment trajectory per course

Rolling-Origin Forecast Model Tournament — STAGE 2 · TOURNAMENT
Seasonal Naive, Damped Trend, Pooled Ridge · 15 origins pick lowest mean absolute error (MAE)

↓ 600 out-of-sample backtest residuals + point forecasts

Split Conformal Interval Calibration — CORE CONTRIBUTION · STAGE 3
Distribution-free per-course 80% & 95% upper bounds (34/40 & 39/40 empirical coverage)

↓ Calibrated 80% Upper Demand Bound

Integer Programming Course & Room Scheduler — STAGE 4 · OR-TOOLS CP-SAT

- HARD CONSTRAINTS (0 VIOLATIONS): Room cap ≥ 80% bound · 0 overlaps · load limits
- SOFT OBJECTIVE (PENALTY MINIMIZATION): 3-tier prefs: 40% Day, 40% Slot, 20% Room

↓ Conflict-free master schedule + solver telemetry

Coordinator Workspace & What-If Simulator — STAGE 5 · APPLICATION
FastAPI + React + TypeScript · Interactive master schedule & counterfactual dry runs

03 DATA SCOPE: REAL VS. SIMULATED — FOIA Replay

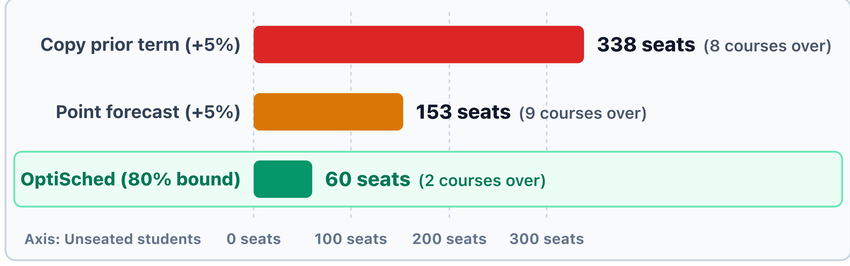
REAL UIUC REGISTRAR DATA
- **40 STEM courses** (22,991 sections)
- **19 training terms** (2016 to 2025)
- **Spring 2026** held-out actuals
- **Historical slot prefs** (25 courses)

SIMULATED SCHEDULING POOL
- **16-room pool** (68 to 322 seats)
- **3-tier prefs** (Day/Slot/Room)
- **3 What-If shocks** (surge/buyout)
- Evaluates solver room packing

Forecasting Horizon: 19 historical terms (ten academic years) train the models to predict **one semester ahead** (Spring 2026), not ten years out.

04 EMPIRICAL RESULTS ON HELD-OUT SPRING 2026 — 40 STEM Courses

A. UNSEATED STUDENT SEATS & OVERFLOWS — Lower is better · Baselines sized at +5%



B. POINT FORECAST MEAN ABSOLUTE ERROR (MAE) — Conformal coverage (40 courses)

28.25 MAE (seats) — Copy Prior Term (1-term)

18.03 −36% vs. prior — Seasonal Naive

17.24 −39% vs. prior — Pooled Ridge (−4% vs. naive)

80% Bound: 34 / 40 **85.0% Coverage**

95% Bound: 39 / 40 **97.5% Coverage**

Key Finding: Pooled Ridge reduces point MAE by 4% over Seasonal Naive (18.03 → 17.24 seats), yet point forecasts alone still under-predict 19 of 40 courses. Sizing rooms to the **calibrated 80% conformal upper bound** drives the 61% reduction in unseated students.

C. SCHEDULE QUALITY VS. ACTUAL UIUC REGISTRAR — Held-out Spring 2026

SCHEDULE METRIC	OPTISCHED	ACTUAL UIUC	OPERATIONAL TAKEAWAY
Seat utilization (enrolled / cap)	64.9%	64.5%	Matches registrar efficiency (+0.4 pp)
Empty-seat waste (unused cap)	35.1%	36.6%	1.5 pp less wasted classroom capacity
Midday 10am to 2pm share	22.5%	47.9%	Spreads peak midday classroom load
Faculty preference match	84.0% (21/25)	n/a	3-tier score (40% Day, 40% Slot, 20% Room)

D. OPERATIONAL IMPACT BY STAKEHOLDER — Students · Faculty · Administrators

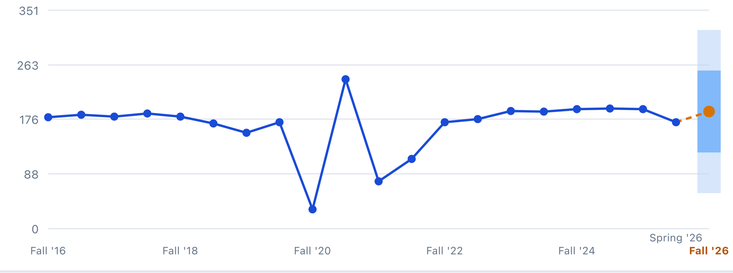
For Students
Cuts week-one unseated seats from 153 to 60 across 40 STEM courses, preventing registration lockouts in high-variance classes.

For Faculty
Honors 3-tier preferences (40% Day, 40% Slot, 20% Room), matching 84.0% (21/25) of preferred teaching slots with 0 time conflicts.

For Administrators
Reduces overflowed courses from 9 to 2 without over-allocating large halls, keeping 64.9% utilization across the 16-room pool.

05 LIVE APPLICATION DEMONSTRATION — FastAPI + React UI

STAT 400 Demand Forecast & Conformal Bounds — FALL 2026



Forward Projection: Historical trajectory (2016 to 2026) and Fall 2026 forecast for STAT 400 with 188-seat point forecast, 80% interval (122 to 254), and 95% interval (57 to 319).

CP-SAT Master Schedule (40 STEM Sections) — 0 CONFLICTS

COURSE	MEETING DAYS & TIME	CLASSROOM	INSTRUCTOR
PHYS 101 — PHYS 101 College Physics: Mech & Heat	MW 08:00 – 09:15	LL 141	Dr. Katherine Ansell
ECE 342 — ECE 342 Electronic Circuits	MW 09:30 – 10:45	ECEB 1013	Dr. Chandrasekhar Radhakrishnan
MATH 441 — MATH 441 Differential Equations	MW 09:30 – 10:45	ECEB 2017	Dr. Nikolaos Tzirakis
CHEM 315 — CHEM 315 Instrumental Chem Systems Lab	MW 12:30 – 13:45	NL 204	Dr. Andrew Gewirth
MATH 112 — MATH 112 Algebra	MW 12:30 – 13:45	LL 141	Dr. Laura Placzek
PHYS 102 — PHYS 102 College Physics: E&M & Modern	MW 12:30 – 13:45	MSEB 100	Dr. Katherine Ansell
CS 357 — CS 357 Numerical Methods I	MW 14:00 – 15:15	LL 151	Dr. Andreas Kloeckner

Generated Master Schedule: First 7 of 40 scheduled STEM sections showing assigned classroom, capacity sized to the 80% demand bound, time slot, and instructor.

06 WHAT-IF DISRUPTION BENCHMARKS — Dry-Run Re-Solves

1. Demand Surge (CS +20%, STAT +15%) — Best possible schedule found, 0 conflicts
CS 425 relocates to a larger lecture hall to absorb the enrollment surge; 39 of 40 sections remain in their assigned room and time slot.

2. Instructor Unavailable (Grant Buyout) — Best possible schedule found, 0 conflicts
Only STAT 420 is reassigned to a qualified colleague; zero cascading room or time shifts across the other 39 courses.

3. Room Capacity Reduced (×0.9 Shock) — Best possible schedule found, 0 conflicts
Up to 2 sections move to compatible classrooms to absorb the 10% capacity reduction; 38 of 40 sections stay unchanged.

Reproduce: `pytest backend/tests/test_uiuc_foia_replay.py` **Stack:** FastAPI · OR-Tools CP-SAT · React + TypeScript · **353 Automated Tests** (196 backend + 157 frontend) **Extension:** Faculty Commercialization Advisor (26-field UM-16C drafter + 425-passage BM25)